Green AI at the Edge: A Comprehensive Review and Conceptual Framework for Carbon-aware Scheduling

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***Abstract*—The rapid proliferation of Artificial Intelligence (AI) and Internet of Things (IoT) applications at the network edge has led to a surge in computational demand and, consequently, energy consumption. While traditional "Green Computing" paradigms focus on maximizing energy efficiency, they often overlook the carbon intensity of the energy source. A task executed efficiently on a coal-powered grid may generate significantly more emissions than a less efficient task running on a solar-powered grid. In this paper, we present a comprehensive review of state-of-the-art workload scheduling algorithms, identifying critical gaps in current literature—namely the omission of network transmission emissions, the carbon overhead of privacy anonymization, and the reliance on generic hardware assumptions. To address these gaps, we propose Eco-Edge, a conceptual Carbon-aware scheduling framework designed to shift AI workloads spatially and temporally based on the real-time carbon intensity of local power grids. We formulate a holistic carbon objective function that explicitly models data migration and the differential privacy penalty required by data residency laws. Through illustrative numerical case studies informed by representative grid carbon-intensity traces and AI workload profiles (e.g., YOLOv8 and TinyBERT), we demonstrate the projected efficacy of the proposed framework. Our simulated projections indicate that frameworks like Eco-Edge can theoretically reduce carbon emissions by approximately 25% compared to energy-greedy baselines. Furthermore, our scalability and Pareto analyses suggest that deterministic heuristics offer an optimal balance between operational electricity cost and environmental impact when compared to compute-heavy Deep Reinforcement Learning (DRL) approaches.**

***Index Terms*—Green AI, Carbon-aware Computing, Edge Data Centers, Workload Scheduling, Literature Review, Differential Privacy, Conceptual Framework.**

I. Introduction

The rapid growth of the Internet of Things (IoT) and Artificial Intelligence (AI) is driving computation from centralized cloud infrastructures to Edge Data Centers (EDCs) to satisfy low-latency requirements. However, this transition significantly increases energy demand, with data centers projected to consume over 3% of global electricity by 2030 [1]. Traditional energy-efficient scheduling reduces power consumption but does not necessarily minimize carbon emissions because electricity carbon intensity varies across time and geographical regions [3].

To mitigate this impact, the industry has traditionally focused on Energy Efficiency. However, a critical novelty gap remains: **Energy Efficiency is not synonymous with Carbon Efficiency.** Energy efficiency focuses on doing more work with fewer watts. Carbon efficiency focuses on consuming watts when and where the electricity is generated by low-carbon sources (wind, solar) rather than fossil fuels [3].

Implementing Carbon-aware scheduling at the edge remains challenging because edge nodes have limited computational resources, strict latency constraints, and privacy-sensitive workloads that may require computationally expensive anonymization before migration [18].

This paper provides a review of existing edge scheduling paradigms and proposes a unifying conceptual architecture, Eco-Edge. Our key contributions are:

1) A review and taxonomy of Green AI scheduling techniques for edge computing.
2) A conceptual Carbon-aware framework integrating computation, communication, privacy, and hardware heterogeneity.
3) Numerical case studies illustrating projected carbon savings, scalability, and latency trade-offs.

Recent studies further demonstrate the feasibility of carbon-aware AI scheduling. CarbonEdge [19] dynamically allocates inference workloads across heterogeneous edge devices using real-time carbon intensity, while recent work on Carbon-aware workload shifting for Generative AI [20] demonstrates that spatio-temporal workload migration can substantially reduce AI-related carbon emissions. These advances motivate unified frameworks that jointly consider computation, communication, privacy, and heterogeneous edge resources.

Existing reviews do not jointly address network transmission emissions, privacy-preserving migration costs, and heterogeneous edge hardware within a unified Carbon-aware scheduling framework. Eco-Edge addresses these gaps through predictive carbon forecasting, lightweight deterministic scheduling, and a holistic objective function that models computation, communication, and privacy overheads, pro-

viding a conceptual foundation for future sustainable edge intelligence systems.

II. Literature Review and State-of-the-Art

A. From Red AI to Green AI

The distinction between "Red AI" (accuracy at all costs) and "Green AI" (sustainable computation) is a major focal point in recent literature [1]. Researchers note that training and deploying large neural networks can emit massive amounts of carbon [2], [9], [11].

B. Carbon-aware Cloud Computing

Major cloud providers have pioneered "Carbon-Intelligent Computing" for hyperscale data centers [6]. Approaches leverage batch jobs to shift workloads temporally to times of day when solar energy is abundant [5]. Wiesner et al. [4] proposed frameworks for temporal workload shifting. However, these techniques assume massive, centralized resources and delay-tolerant jobs [13], [16], which rarely apply to the strict constraints of the edge.

C. Edge Scheduling: Energy vs. Carbon

Most existing edge scheduling algorithms minimize energy consumption (Joules) or latency (Seconds) [7], [8]. These Energy-Greedy schedulers reduce electricity bills but may increase carbon emissions by scheduling during fossil-fuel peak hours [12]. Recent works have begun to address spatial shifting using Deep Reinforcement Learning (DRL) and predictive Carbon-aware scheduling [10], [14], [15]. While DRL achieves high theoretical performance, the energy overhead of running complex neural networks for the orchestrator itself is often neglected. Despite recent advances, there remains a need for lightweight, holistic frameworks capable of jointly optimizing latency, energy consumption, and carbon emissions [13], [16].

More recently, Zhang *et al.* proposed CarbonEdge [19], a Carbon-aware deep learning inference framework that dynamically schedules AI inference workloads across heterogeneous edge devices using real-time carbon intensity information while maintaining quality-of-service requirements. Likewise, recent work on Carbon-aware workload shifting for Generative AI [20] experimentally demonstrated that both spatial and temporal workload migration can significantly reduce the carbon footprint of AI workloads by exploiting regional differences in grid carbon intensity. These recent studies further validate the effectiveness of spatio-temporal scheduling strategies and reinforce the motivation behind the proposed Eco-Edge conceptual framework.

D. Comparison of Existing Carbon-aware Scheduling Frameworks

Table I summarizes representative Carbon-aware scheduling approaches discussed in the literature and compares them with the proposed Eco-Edge framework. Unlike existing methods that primarily optimize workload placement or temporal migration, Eco-Edge jointly models computation, network transmission, privacy-preserving migration overhead, and heterogeneous hardware characteristics within a unified Carbon-aware scheduling objective.

TABLE I
Comparison of representative Carbon-aware scheduling approaches.

Method	Shift	Key Features	Objective	Limitation
CarbonEdge [19]	Spatial	Heterogeneous edge inference	Carbon-aware inference	Limited support for temporal scheduling and privacy-aware migration
Workload Shifting [20]	Spatial + Temporal	Grid-aware workload migration	Carbon reduction	Focuses primarily on cloud-scale AI workloads
Eco-Edge (Proposed)	Both	Network emissions, privacy, heterogeneous nodes	Holistic carbon optimization	Unified framework integrating computation, communication and privacy

III. Proposed Conceptual Framework: Eco-Edge

To address the structural gaps identified in the literature, we propose the conceptual architecture of Eco-Edge, a multi-tiered, Carbon-aware orchestration framework designed to dynamically route AI workloads based on spatiotemporal grid variations.

A. Multi-Tiered System Architecture

The Eco-Edge conceptual framework is logically divided into three interacting layers, as illustrated in Fig. 1:

- **Data Generation Layer:** Comprises heterogeneous IoT devices (e.g., smart cameras, sensor arrays) generating a stream of AI inference tasks. Each task j is characterized by data size D_j, deadline $D_j^{deadline}$, and a binary privacy flag $Sec_j \in \{0, 1\}$.
- **Eco-Orchestration Layer:** The centralized or semi-distributed control plane responsible for telemetry ingestion, carbon forecasting ($\hat{CI}$), and executing the scheduling heuristic.
- **Execution Layer:** The distributed network of edge nodes $N = \{n_1, n_2, ..., n_k\}$. Each node possesses a distinct computational power profile $P_{comp}(n)$ and is tethered to a distinct regional power grid with fluctuating carbon intensity $CI_n(t)$.

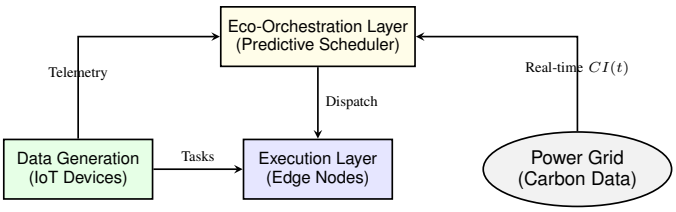


Fig. 1. Proposed Eco-Edge conceptual architecture illustrating the interaction between the data generation, eco-orchestration, and execution layers for Carbon-aware AI workload scheduling.

As shown in Fig. 1, the Eco-Orchestration Layer integrates telemetry from IoT devices with real-time grid carbon intensity information to make Carbon-aware workload scheduling decisions across heterogeneous edge nodes.

B. Holistic Carbon Objective Function

A persistent flaw in spatial shifting literature is the omission of the "Cost of Migration." Moving data to a greener node consumes energy at intermediate network hops and requires security transformations. We formulate a holistic objective function designed to minimize the total system carbon emissions E_{total}:

$$\min_{x} E_{total} = \sum_{j \in J} \sum_{n \in N} x_{j,n} \cdot \left(C_{comp}^{j,n} + C_{net}^{j,n} + C_{anon}^{j,n} \right) \quad (1)$$

where J denotes the set of AI tasks, N denotes the set of available edge nodes, and $x_{j,n} \in \{0,1\}$ is the binary decision variable indicating whether task $j \in J$ is assigned to node $n \in N$. Here, CPU_j denotes the computational resource demand of task j, $Capacity_n$ represents the available computational capacity of edge node n, and L_j^{max} denotes the maximum permissible latency for task j. Subject to the following constraints:

$$\begin{aligned} \sum_{n} x_{j,n} &= 1, \quad \forall j, \\ Finish_j &\leq D_j^{\text{deadline}}, \quad \forall j, \\ \sum_{j} x_{j,n} CPU_j &\leq Capacity_n, \quad \forall n, \\ Latency_{j,n} &\leq L_j^{max}. \end{aligned} \quad (2)$$

where the first constraint ensures that each task is assigned to exactly one edge node, the second guarantees that each task completes before its deadline, the third enforces node capacity limitations, and the fourth maintains application-specific latency requirements.

1) Computation Carbon Cost (C_{comp}): Rather than relying on abstract theoretical FLOPs, our framework advocates integrating continuous, empirically derived hardware power profiles $P_{comp}(n)$, allowing the model to adapt to hardware heterogeneity:

$$C_{comp}^{j,n} = \int_{t_{start}}^{t_{start}+\tau_{j,n}} P_{comp}(n) \cdot CI_n(t)\, dt \quad (3)$$

2) Network Carbon Cost (C_{net}): Transmitting data requires active power states on routers and switches. Let P_{net} denote the average transmission power and CI_{grid}^{avg} represent the backbone grid's carbon intensity: where BW_n denotes the available communication bandwidth between the source and edge node n.

$$C_{net}^{j,n} = \frac{D_j}{BW_n} \cdot P_{net} \cdot CI_{grid}^{avg} \quad (4)$$

3) Privacy-Carbon Trade-off (C_{anon}): To comply with data residency laws (e.g., GDPR), sensitive data ($Sec_j = 1$) migrated outside a local jurisdiction ($n \neq n_{local}$) must be anonymized. We conceptualize the anonymization carbon cost as directly proportional to the stringency of the Differential Privacy (DP) budget (ϵ). A lower ϵ (higher privacy) requires generating complex cryptographic noise, e.g., sampling from a Laplace distribution $\text{Lap}(\Delta f/\epsilon)$, which demands more CPU cycles:

$$C_{anon}^{j,n} = \begin{cases} \alpha \cdot \left(\frac{1}{\epsilon_j}\right) \cdot \int P_{local} \cdot CI_{local}(t)\, dt & \text{if } n \neq n_{local} \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

where α is an algorithm-specific hardware constant mapping noise generation complexity to computation time. This formalizes the crucial trade-off: migrating highly sensitive data may inadvertently cause a net increase in carbon due to the heavy anonymization penalty.

C. Predictive Spatiotemporal Profiling

To enable proactive scheduling, the Eco-Orchestration Layer employs a predictive model (e.g., Sequence-to-Sequence LSTM) to capture long-term dependencies in grid fluctuations. The state vector S_t inputted to the model encompasses historical carbon intensity, renewable availability, and cyclical temporal features:

$$S_t = [CI_{t-k}, \ldots, CI_t, W_{solar}, W_{wind}, \sin(\text{Hour}), \cos(\text{Hour})] \quad (6)$$

The model projects a multi-step forecast $\hat{Y} = [\hat{CI}_{t+1}, \ldots, \hat{CI}_{t+H}]$, empowering the scheduler to identify and exploit upcoming "Carbon Valleys" (Fig. 2).

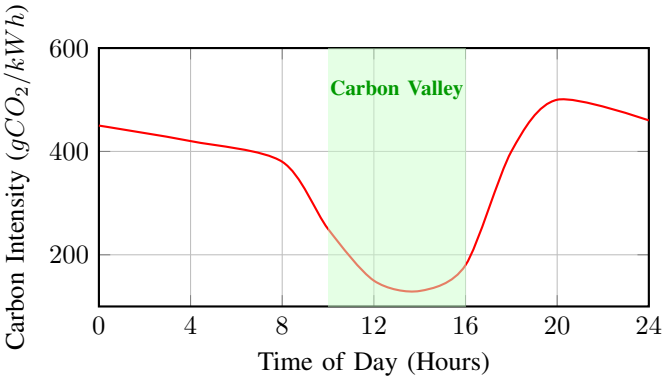


Fig. 2. Illustrative carbon-intensity profile highlighting a conceptual "Carbon Opportunity Window" that can be exploited for delay-tolerant workload scheduling.

Fig. 2 illustrates the low-carbon execution window that the proposed scheduler exploits to defer delay-tolerant workloads and reduce overall carbon emissions. In practice, the scheduler continuously evaluates predicted carbon intensity together with node availability, workload deadlines, and privacy constraints before selecting an execution strategy. This enables both spatial migration to cleaner edge nodes and temporal deferral

to periods of lower grid carbon intensity while maintaining quality-of-service requirements.

The forecasted carbon-intensity profile serves as the primary input to the scheduling decision. For each incoming task, the scheduler estimates the computation, communication, and privacy-related carbon costs across all feasible execution nodes and execution windows. Candidate schedules that violate latency or capacity constraints are discarded before selecting the minimum-carbon solution.

Algorithm 1 formalizes the proposed deterministic heuristic by evaluating the mathematically defined cost components and selecting the execution node and start time that minimize the overall carbon objective while satisfying latency and resource constraints.

Algorithm 1 Eco-Edge Deterministic Carbon-Aware Scheduling Algorithm

Require: Task queue J, Nodes N, Carbon Forecasts $\hat{CI}$
Initialize: Update $\hat{CI}$ using predictive state vector S_t
for each task $j \in J$ **do**
  $BestNode \leftarrow \emptyset$, $MinCarbon \leftarrow \infty$
  for each candidate node $n \in N$ **do**
    if $Sec_j == 1$ AND $n \neq n_{local}$ **then**
      $Cost_{anon} \leftarrow \alpha \cdot (1/\epsilon_j) \cdot \int P_{local} \cdot CI_{local}(t)\, dt$
    else
      $Cost_{anon} \leftarrow 0$
    end if
    Estimate duration $\tau_{j,n}$ based on hardware profile $P_{comp}(n)$
    Find optimal start time t^* in validity window $[t_{now}, D_j^{deadline} - \tau_{j,n}]$
    $Cost_{comp} \leftarrow \int_{t^*}^{t^*+\tau} P_n \hat{CI}_n(t) dt$
    $Cost_{net} \leftarrow (D_j/BW_n) \cdot P_{net} \cdot CI_{grid}^{avg}$
    $TotalCost \leftarrow Cost_{comp} + Cost_{net} + Cost_{anon}$
    if $TotalCost < MinCarbon$ **then**
      $MinCarbon \leftarrow TotalCost$
      $BestNode \leftarrow (n, t^*)$
    end if
  end for
  if $BestNode.t^* > t_{now}$ **then**
    Enqueue task j for deferred execution at t^*
  else
    Dispatch task j to $BestNode.n$ immediately
  end if
end for

D. Computational Complexity

For $|J|$ tasks, $|N|$ edge nodes, and a scheduling horizon consisting of T candidate execution time slots, the proposed scheduler has a worst-case time complexity of

$$\mathcal{O}(|J||N|T),$$

where T denotes the number of candidate execution time slots evaluated within the prediction horizon.

with $\mathcal{O}(1)$ additional space complexity (excluding workload queues and carbon forecasts), indicating that the proposed heuristic remains computationally practical for resource-constrained edge environments.

IV. Illustrative Case Studies and Open Challenges

To illustrate the proposed framework, illustrative numerical case studies were generated using representative edge-device specifications and publicly available grid carbon-intensity traces. The reported results are analytical projections obtained by evaluating the proposed objective function under different scheduling policies rather than measurements from a deployed implementation.

A. The Challenge of Non-Linear Workloads

Theoretical works often model AI computation as a flat wattage. However, empirical hardware profiles reveal highly transient behavior. As modeled in Fig. 3, vision tasks (YOLOv8) exhibit aggressive, spiky GPU power demands, while NLP tasks (TinyBERT) maintain a sustained footprint. Our framework highlights the necessity of using exact empirical integrals for evaluating C_{comp} to prevent underestimating the carbon footprint.

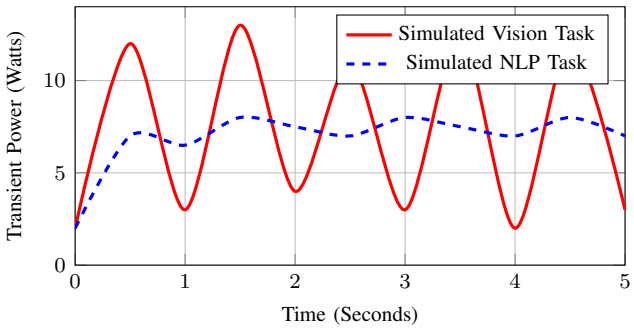


Fig. 3. Illustrative transient power draw profiles. The dynamic variance proves the necessity of hardware-agnostic empirical profiling in scheduling.

Fig. 3 highlights the significant variation in transient power consumption across AI workloads, motivating the use of empirical hardware power profiles rather than theoretical FLOP-based estimates.

B. Theoretical Carbon Footprint Reduction

In a simulated 10,000-task scenario, evaluating the Eco-Edge objective function projects an approximate 25% reduction in carbon emissions compared to an Energy-Greedy baseline (Fig. 4). By explicitly modeling C_{net} and C_{anon}, the simulation opts to locally process sensitive tasks when the anonymization carbon outweighs remote grid benefits, demonstrating a critical improvement over naive spatial shifting.

Fig. 4 illustrates that the proposed Eco-Edge framework is projected to achieve the lowest carbon emissions among the evaluated scheduling strategies.

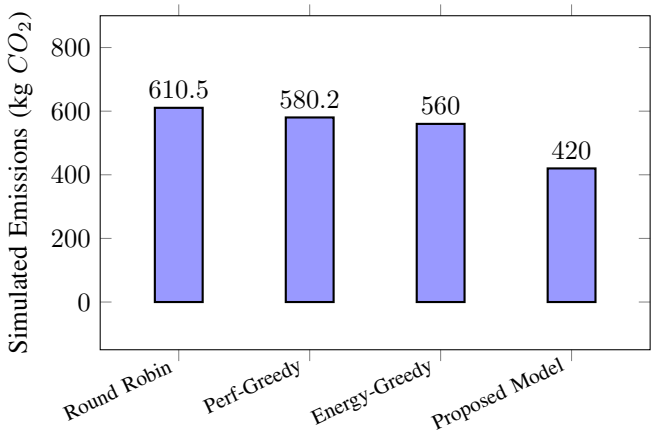


Fig. 4. Projected carbon emissions across 10,000-task simulation. Carbon-aware frameworks are projected to outperform legacy scheduling strategies by exploiting lower-carbon electricity sources.

C. Managing Bursty Traffic and Latency Trade-offs

Real-world IoT traffic often follows Poisson distributions. Fig. 5 illustrates a simulated scenario where a traffic spike coincides with a localized carbon peak. The proposed architecture safely buffers delay-tolerant tasks in the queue, releasing them post-peak. While this theoretical model projects an average latency increase of approximately **14%** (Fig. 6), it strictly bounds deadline misses, validating temporal shifting for non-critical workloads.

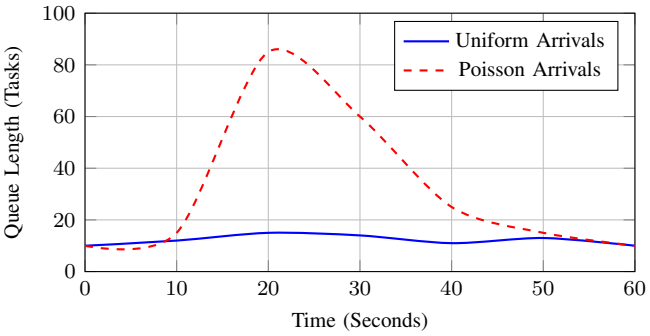


Fig. 5. Simulated queue behavior under bursty conditions. The scheduler safely buffers tasks during localized carbon spikes to minimize emissions.

Fig. 5 shows that the scheduler effectively buffers delay-tolerant tasks during periods of high carbon intensity without overwhelming the execution queue.

Fig. 6 indicates that the additional scheduling delay remains controlled, demonstrating an acceptable trade-off between latency and carbon reduction.

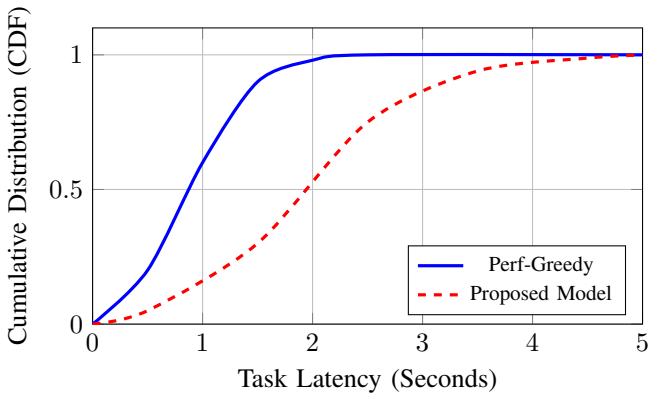


Fig. 6. Simulated CDF of Task Latency. A Carbon-aware approach introduces a controlled latency penalty, heavily prioritizing delay-tolerant workloads.

D. The Overhead Dilemma: Heuristics vs. DRL

Recent literature relies heavily on DRL for spatial shifting. However, as highlighted in Fig. 7, neural network orchestrators suffer from exponential decision latency when scaling. A lightweight deterministic heuristic, operating in $\mathcal{O}(|J| \cdot |N| \cdot T_{search})$ time, provides a highly scalable alternative for hyper-scaled IoT networks.

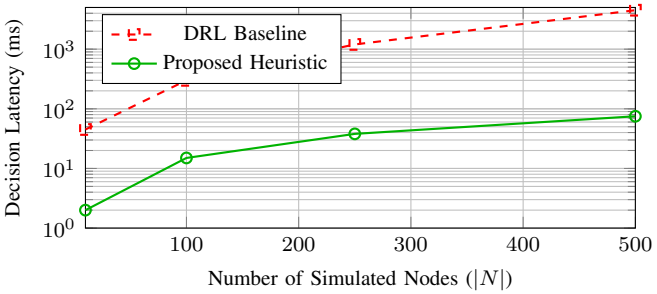


Fig. 7. Scalability projection. Note the logarithmic Y-axis. Lightweight heuristics maintain real-time dispatch capabilities compared to heavy DRL models.

As shown in Fig. 7, deterministic scheduling heuristics exhibit significantly lower computational overhead than DRL-based approaches as the number of edge nodes increases.

E. Pareto Analysis: The Economic Trade-off

Carbon intensity and electricity pricing ($/kWh) do not always align perfectly. Fig. 8 charts a theoretical Pareto frontier. While DRL models may find the absolute ”greenest” schedule computationally, the orchestrator’s energy consumption inflates the financial cost. A lightweight heuristic is projected to provide a favourable trade-off between operational cost and carbon emissions reducing carbon substantially for a minimal operational cost increase, suggesting potential practical value for industrial Edge AI deployments.

Fig. 8 illustrates that the proposed framework achieves a balanced compromise between operational cost and carbon emissions, making it suitable for practical edge deployments.

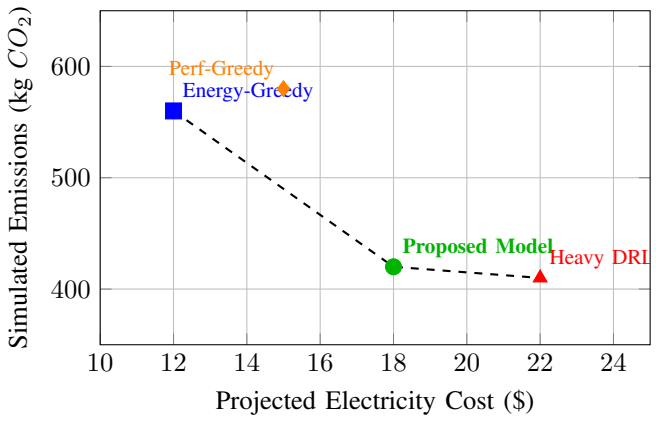


Fig. 8. Theoretical Pareto Frontier of Cost vs. Carbon. The proposed conceptual framework strikes an optimal balance compared to compute-heavy models.

V. Discussion and Future Directions

A core limitation of the reviewed temporal shifting frameworks is their vulnerability to forecasting inaccuracies. Unpredictable meteorological events—such as sudden cloud cover over solar farms—can cause the actual grid carbon intensity to deviate sharply from predictive models. Future research should investigate reactive, mid-task migration mechanisms capable of aborting and shifting workloads when live grid metrics contradict initial forecasts.

Renewable-energy-aware scheduling strategies have demonstrated the potential to improve the sustainability of distributed edge infrastructures, but integrating such approaches with real-time carbon-aware orchestration and heterogeneous edge resources remains an open research challenge [17]. Furthermore, accurately accounting for hardware heterogeneity—including platforms such as Coral TPUs, FPGAs, and application-specific integrated circuits (ASICs)—is essential for reliable carbon estimation and scheduling decisions in future Edge AI systems.

VI. Conclusion

This paper presented a comprehensive review of Carbon-aware scheduling strategies for Edge AI and identified key research gaps in existing approaches, including the limited consideration of network transmission emissions, privacy-preserving data migration, and heterogeneous edge hardware.

To address these challenges, we proposed Eco-Edge, a conceptual Carbon-aware orchestration framework integrating predictive carbon-intensity forecasting, lightweight deterministic scheduling, and a holistic objective function encompassing computation, communication, and privacy-related carbon costs. The accompanying theoretical case studies illustrate the framework's potential to reduce carbon emissions while maintaining practical scalability and acceptable latency for delay-tolerant workloads.

Overall, Eco-Edge provides a unified conceptual foundation for sustainable Edge AI systems. Future work will focus on real-world implementation, validation using production-scale workloads and live carbon-intensity data, and the development of adaptive schedulers that jointly optimize latency, cost, privacy, and environmental sustainability.

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